

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set(style='whitegrid')
```

```
In [3]: df = pd.read_csv('train.csv')
df.head()
```

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style='whitegrid')

df = pd.read_csv('train.csv')
df.head()
```

```
Out[3]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Hekkinen, Miss. Laina	female	26.0	0	0	STON/OZ. 3101282	7.9250	NaN	S
3	4	1	1	Futelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

```
In [4]: df.shape
```

```
Out[4]: (891, 12)
```

```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column             Non-Null Count  Dtype
---  --
0   PassengerId         891 non-null    int64
1   Survived            891 non-null    int64
2   Pclass              891 non-null    int64
3   Name                891 non-null    object
4   Sex                 891 non-null    object
5   Age                 714 non-null    float64
6   SibSp               891 non-null    int64
7   Parch               891 non-null    int64
8   Ticket              891 non-null    object
9   Fare                891 non-null    float64
10  Cabin               204 non-null    object
11  Embarked            889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [6]: df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.69118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526487	1.102743	0.809057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [7]: df.isnull().sum()
```

```
Out[7]: PassengerId    0
Survived              0
Pclass                0
Name                  0
Sex                   0
Age                  177
SibSp                 0
Parch                 0
Ticket                0
Fare                  0
Cabin                 687
Embarked              2
dtype: int64
```

```
In [8]: df['Sex'].value_counts()
df['Embarked'].value_counts()
```

```
Out[8]: Embarked
S    444
C    168
Q     77
Name: count, dtype: int64
```

```
In [9]: df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

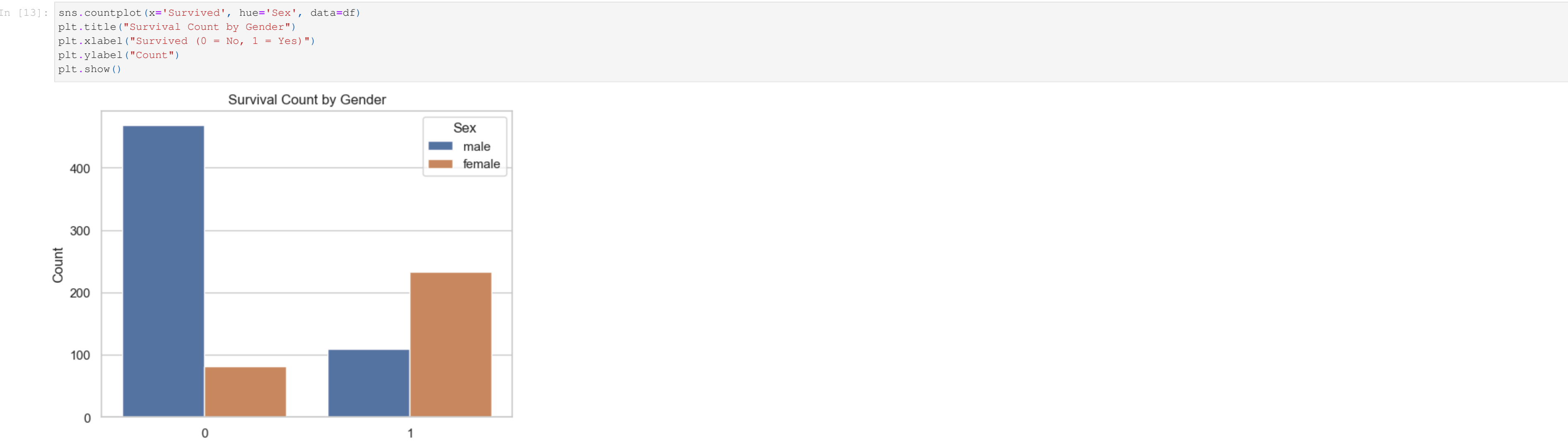
```
In [10]: df.drop('Cabin', axis=1, inplace=True)
```

```
In [11]: df['Age'].fillna(df['Age'].median(), inplace=True)
```

```
In [12]: df.isnull().sum()
```

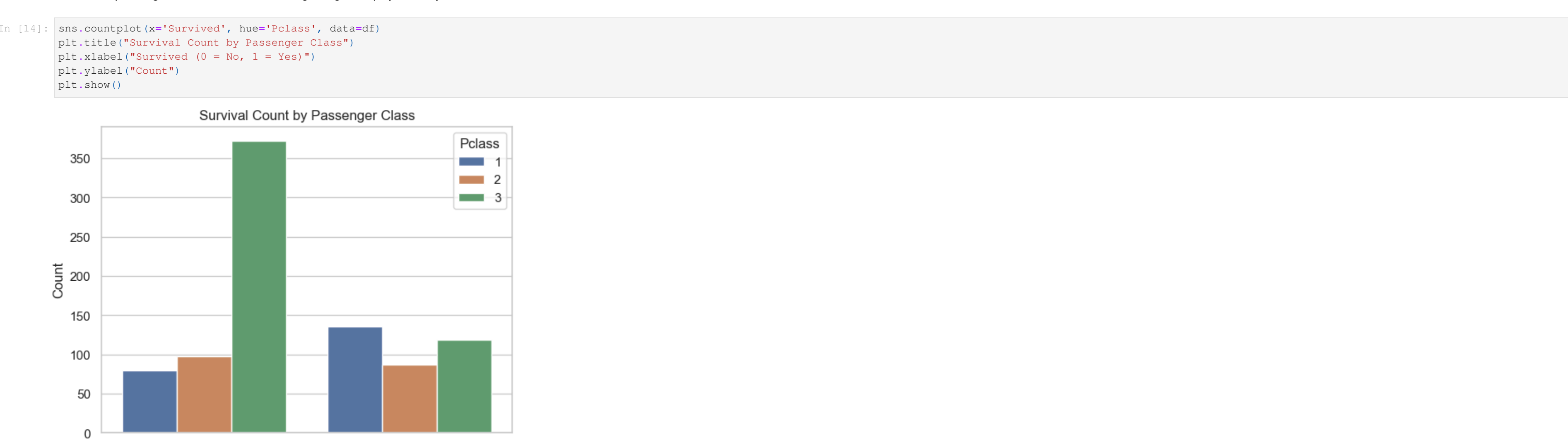
```
Out[12]: PassengerId    0
Survived              0
Pclass                0
Name                  0
Sex                   0
Age                  177
SibSp                 0
Parch                 0
Ticket                0
Fare                  0
Cabin                 0
Embarked              0
dtype: int64
```

```
In [13]: sns.countplot(x='Survived', hue='Sex', data=df)
plt.title('Survival Count by Gender')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Count')
plt.show()
```



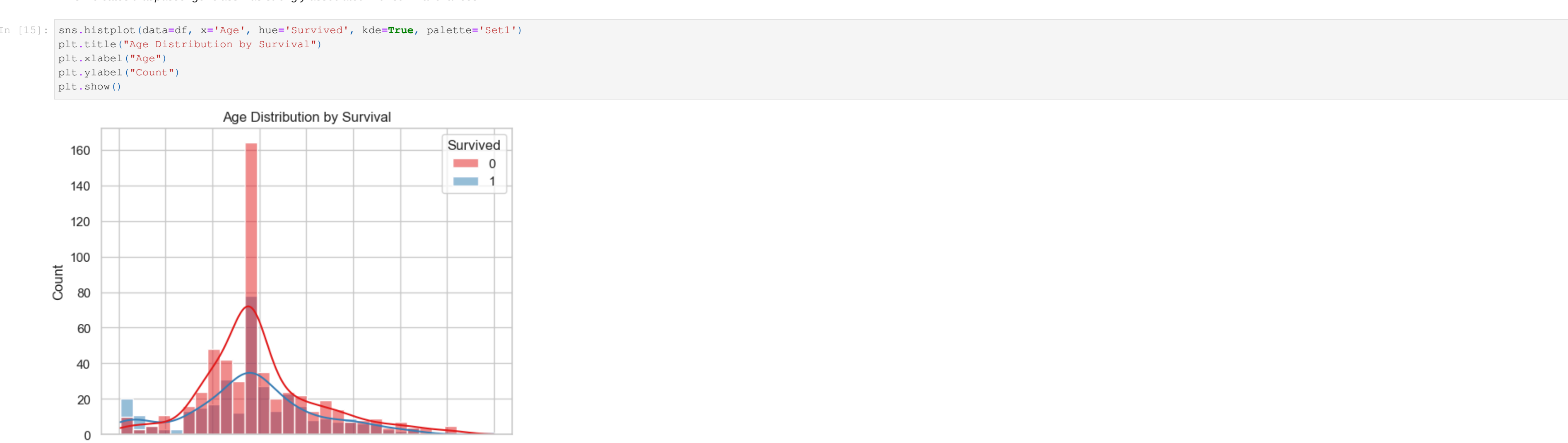
Observation:
The count plot shows that a significantly higher proportion of females survived compared to males.
Most male passengers did not survive, indicating that gender played a major role in survival outcome.

```
In [14]: sns.countplot(x='Survived', hue='Pclass', data=df)
plt.title('Survival Count by Passenger Class')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Count')
plt.show()
```



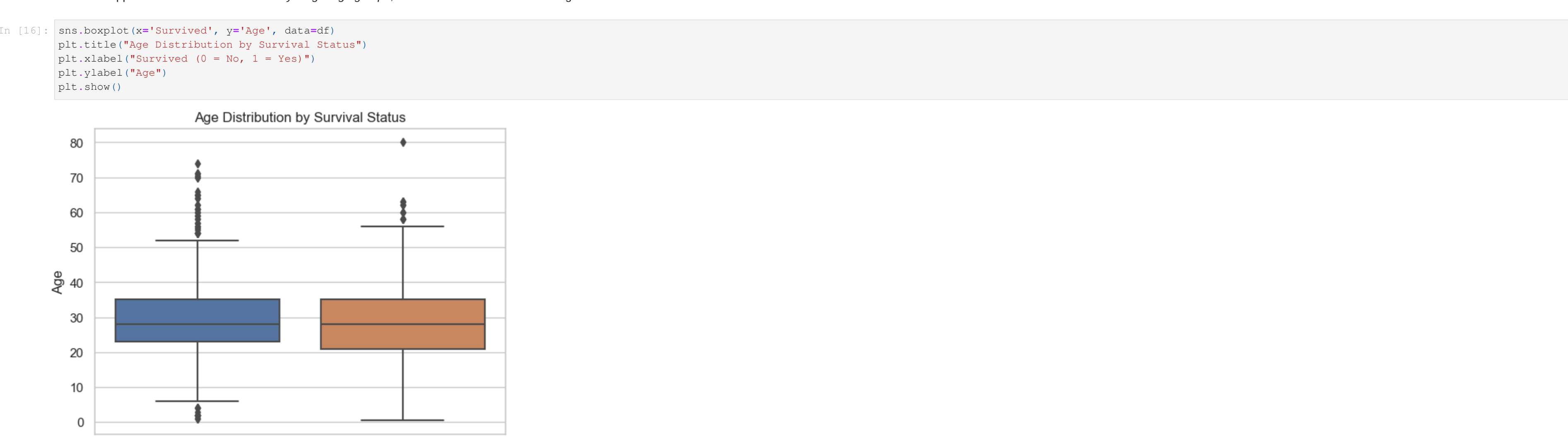
Observation:
Passengers in 1st class had the highest survival rate, while those in 3rd class had the lowest.
This indicates that passenger class was strongly associated with survival chances.

```
In [15]: sns.histplot(data=df, x='Age', hue='Survived', kde=True, palette='Set1')
plt.title('Age Distribution by Survival')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()
```



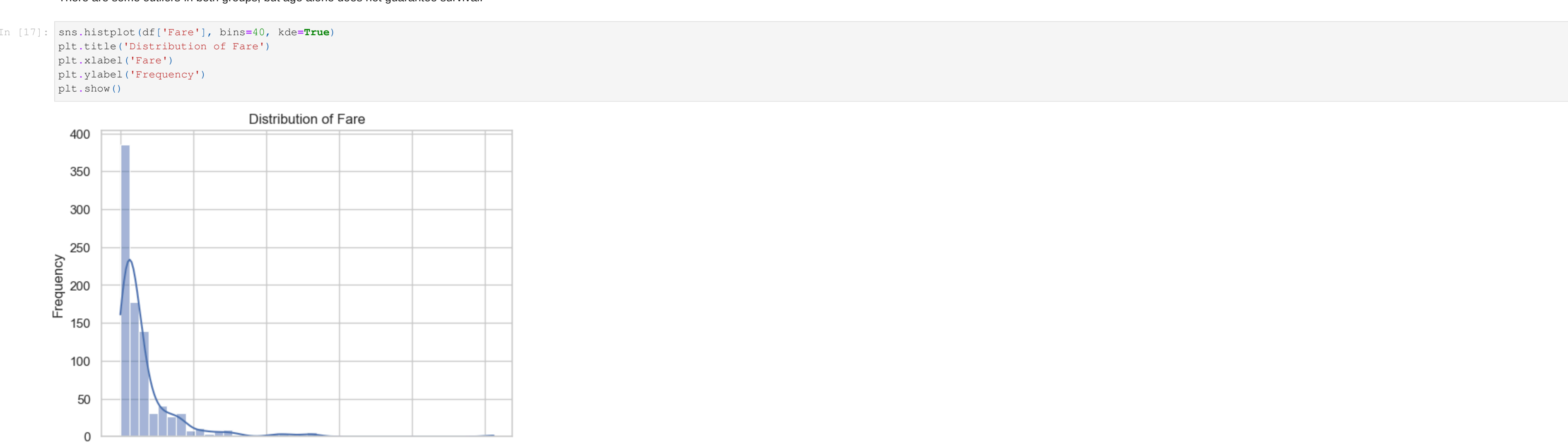
Observation:
Most passengers were between 20-40 years old.
Survivors appear to be more common in the younger age groups, but survival occurred across all ages.

```
In [16]: sns.boxplot(x='Survived', y='Age', data=df)
plt.title('Age Distribution by Survival Status')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Age')
plt.show()
```



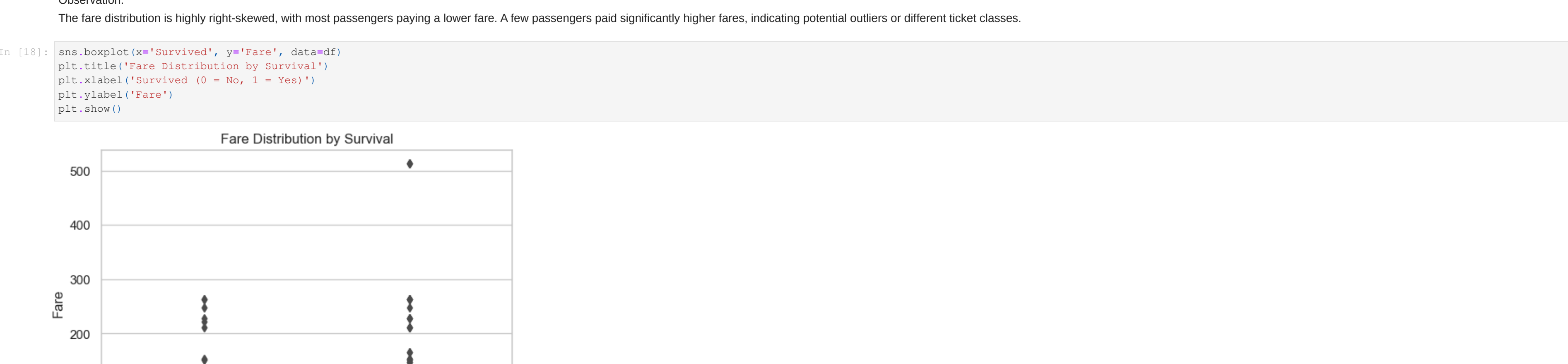
Observation:
The median age of survivors appears to be slightly lower than that of non-survivors.
There are some outliers in both groups, but age alone does not guarantee survival.

```
In [17]: sns.histplot(df['Fare'], bins=40, kde=True)
plt.title('Distribution of Fare')
plt.xlabel('Fare')
plt.ylabel('Frequency')
plt.show()
```



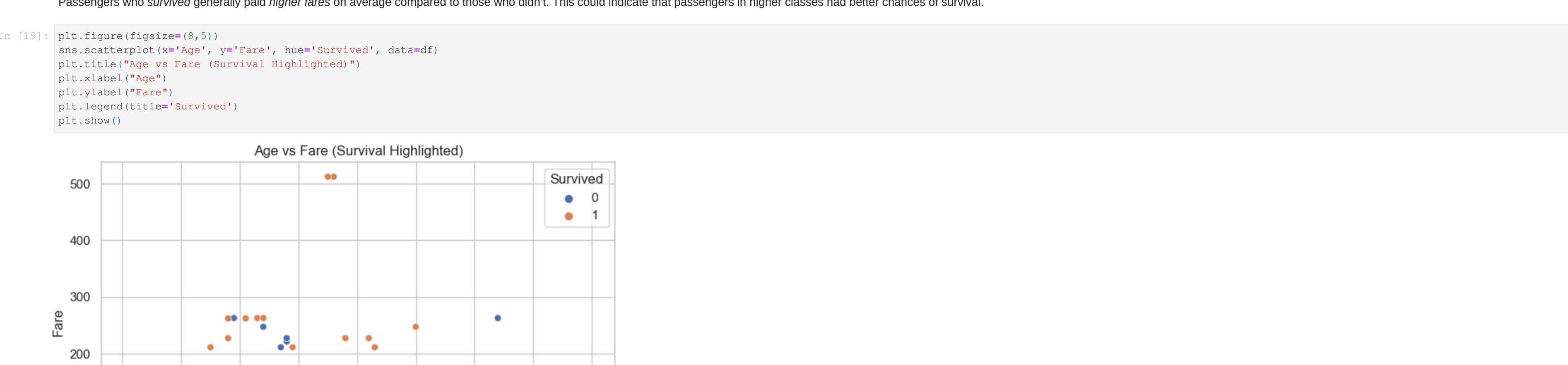
Observation:
The fare distribution is highly right-skewed, with most passengers paying a lower fare. A few passengers paid significantly higher fares, indicating potential outliers or different ticket classes.

```
In [18]: sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Fare')
plt.show()
```



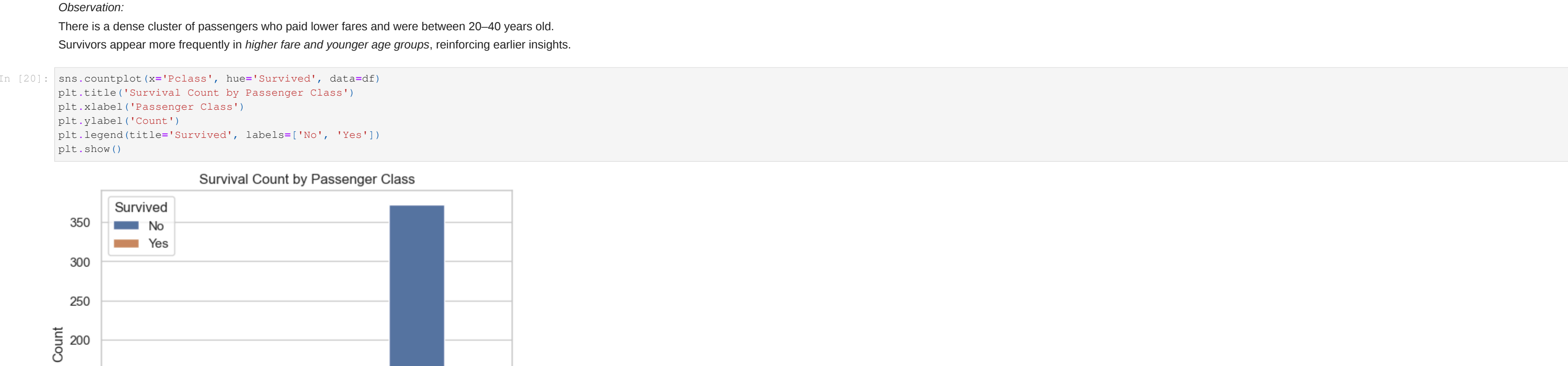
Observation:
Passengers who survived generally paid higher fares on average compared to those who didn't. This could indicate that passengers in higher classes had better chances of survival.

```
In [19]: plt.figure(figsize=(8,5))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title('Age vs Fare (Survival Highlighted)')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.legend(title='Survived')
plt.show()
```



Observation:
There is a dense cluster of passengers who paid lower fares and were between 20-40 years old.
Survivors appear more frequently in higher fare and younger age groups, reinforcing earlier insights.

```
In [20]: sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival Count by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Count')
plt.legend(title='Survived', labels=['No', 'Yes'])
plt.show()
```



Observation:
Passengers in 1st class had the highest survival rate, followed by those in 2nd class, while 3rd class passengers had the lowest. This indicates a strong correlation between socio-economic status and chances of survival.

```
In [21]: sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']])
plt.suptitle('Pairplot of Selected Numerical Features', y=1.02)
plt.show()
```



Observation:
There is a dense cluster of passengers who paid lower fares and were between 20-40 years old.
Survivors appear more frequently in higher fare and younger age groups, reinforcing earlier insights.

```
In [22]: plt.figure(figsize=(10, 6))
sns.heatmap(df.select_dtypes(include='number').corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```



Observation:
The heatmap shows a positive correlation between Fare and Survived, and a negative correlation between Pclass and Survived.
This confirms that higher class and fare are associated with a better chance of survival.

Summary of Findings:

- Gender:** Female passengers had a significantly higher survival rate than males.
- Passenger Class:** 1st class passengers had better survival chances than those in 2nd and 3rd class.
- Fare:** Passengers who paid higher fares were more likely to survive.
- Age:** Most passengers were between 20-40 years old. Survival was slightly higher among younger passengers.
- Embarked:** Most passengers boarded from port 'S', and survival rates varied slightly by embarkation point.

• **Correlations:** Fare and Pclass showed the strongest correlations with survival. Higher fare and lower class number (1st class) indicated a better chance of survival.

This analysis highlights how socio-economic status, gender, and ticket fare played important roles in survival outcomes aboard the Titanic.